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# THE CLIMB SPACE

Medium-Fi Prototype

Team 18

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# ALTROUTE

Different bodies. Different Solutions.

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TEAM MEMBER

# MEET THE TEAM

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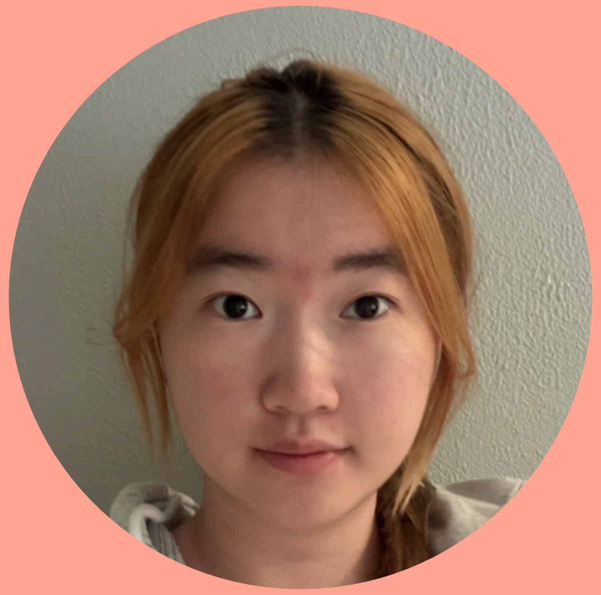

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# PROBLEM & SOLUTION OVERVIEW

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# WHAT'S THE PROBLEM?

Climbers often struggle to find community-shared betas that adequately accomodate their different body types and truly assist them in their climbs.

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# WHAT'S THE SOLUTION?

Provide personalized, AI-generated betas based on each climber's physiques and skill level, prioritzing their safety while maximizing satisfaction by still encouraging problem-solving

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# VALUES IN DESIGN

# VALUES IN DESIGN

INCLUSIVITY

USER AGENCY

ACCESSIBILITY

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# INCLUSIVITY

In-context meaning:
Climbing solutions should adapt to different bodies rather than assuming a single "correct" way to climb. The system supports diverse physiques, abilities, and constraints by generating personalized betas.

## Physique-Aware Personalization

- Users input height, reach, skill level, and physical constraints
- AI generates multiple beta options tailored to body differences
- Removes reliance on "one-size-fits-all" community beta

## Multiple Valid Solutions

- Several beta styles instead of one optimal path
- Encourages climbers to find strategies that fit their own movement style
- Frames difficulty as relational, not absolute

# USER AGENCY

In-context meaning:
The climber controls how much assistance they receive, when to progress, and which strategy to follow. AI supports decision-making rather than replacing it.

Guidance Control

- Users control pacing (“Next,” “Repeat,” “Skip”)
- Users decide when to progress
- AI suggests — user chooses

Choice & Exploration

- Multiple beta options available for selection
- Users can switch strategies anytime
- AI suggests — user decides

Instruction Depth

- Gradual reveal instead of full solution upfront
- Maintains problem-solving and learning experience
- Supports both beginners and experienced climbers

# ACCESSIBILITY

In-context meaning:
The system minimizes barriers to participation by using devices climbers already own and workflows that fit naturally into gym environments.

Low Barrier Hardware
- Designed for smartphones and optional earbuds
- No expensive or specialized equipment required
- Works within existing climbing routines

Flexible Use Modes
- Voice guidance optional
- Visual interaction still fully supported
- Users choose interaction style based on comfort and context

Seamless Gym Integration
- Quick interaction between attempts
- Hands-free audio option reduces need to touch phone with chalked hands
- Does not obstruct vision like AR headsets

# VALUE TENSIONS

## AGENCY VS AI GUIDANCE

- Too much AI guidance can reduce the climber's sense of discovery and ownership of the climb.
- Too little guidance may leave users frustrated when stuck.

Mitigation:
Use step-by-step hint reveal instead of showing the full beta at once. Provide multiple beta options so AI suggests possibilities while users retain decision-making control.

## ACCESSIBILITY VS SAFETY

- Real-time voice coaching improves hands-free accessibility during climbing.
- However, audio instructions may distract climbers or reduce environmental awareness in a busy gym.

Mitigation:
Make voice guidance optional and recommend single-earbud use. Always provide a visual alternative so users can choose safer interaction modes depending on context.

## PERSONALIZATION VS COMMUNITY LEARNING

- Highly personalized AI betas support diverse bodies and individual needs.
- Over-personalization may weaken traditional climbing culture built on shared beta and social learning.

Mitigation:
Combine AI-generated betas with community-shared betas. Allow users to share customized betas and browse others' strategies to preserve collaborative learning.

## CLARITY VS COGNITIVE LOAD

- Detailed explanations help users understand movement strategies.
- Too much information during climbing can overwhelm attention and interrupt flow.

Mitigation:
Present one primary instruction per step. Gradually reveal information only when requested.

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# OUR TASKS

# SIMPLE

**Users can search for a route-specific beta by searching for the gym and filtering the betas.**

This task is simple because it requires the lowest mental and physical effort. Users only need to search up a gym and select the route to find a beta.

# MODERATE

**Users can enter route details, physical attributes, and skill level to generate a personalized beta using AI**

This task is moderate because it requires users to upload an image of the route, select the holds, and enter their details, requiring more effort than the simple task. This stage introduces the app's primary innovation requiring intentional user input and deeper engagement.

# COMPLEX

**Users receive real-time guidance from the AI voice coach or visual guides for specific steps in the climb**

This task is complex because users must divide their attention between climbing movements, environmental awareness, and system guidance, requiring significantly more mental effort than the other tasks.

DIFFICULTY IS BASED ON APP USAGE EFFORT (NOT CLIMBING GRADE).

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# CHANGES IN TASKS

## SIMPLE

Users can find betas for a specific gym

Users can search for a route-specific beta by searching for the gym and filtering the betas.

Based on our usability testing, climbers preferred viewing betas shared by friends or filtering by body type for their current routes rather than relying on generic beta

# CHANGES IN TASKS



## MODERATE

Users can enter route details, physical attributes, and skill level to generate a personalized beta using AI

This task represents the core workflow of our app, and usability testing did not reveal issues that required changes to the task itself.

# CHANGES IN TASKS

## COMPLEX

Users receive real-time guidance from the AI voice coach for a specific step in the climb

⬇

**Users receive real-time guidance from the AI voice coach or visual guides for specific steps in the climb**

Based on our usability testing, we found that some climbers prefer not to wear earphones while climbing. We decided to make this an optional feature, giving users the choice between an AI voice coach or simply viewing the visual guides on their phones.

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# USABILITY GOALS & KEY MEASUREMENTS

# GOALS & MEASUREMENTS

## Efficiency

- Measurement: Time taken to complete each task (seconds):
  - T1: Find & watch a beta
  - T2: Generate a customized beta
  - T3: Complete step-by-step beta

- Why?

In a climbing gym, people check beta between attempts. If the flow takes too long, climbers are unlikely to use it during a session.

## Clarity / Error Prevention

- Measurement: Errors per task (#)

- What we count as an "error"
  - Wrong navigation; Misinterpreting a control; Getting stuck and needing a hint to continue

- Why?

Our app has unique features (physique-based beta + step mode + save/resume); We need to confirm users understand the flow without coaching.

# RESULTS FROM USABILITY TESTING

## Efficiency

- 100% completion across all tasks (3/3 participants)
- Task 1: 0:23–1:08 (fast)
- Task 2: 0:45–2:30 (largest spread)
- Task 3: 1:10–2:14 (moderate)

## Clarity / Error Prevention

- Task 1 errors: (0-1), just one user have one error
- Task 2 had the most errors (0–2)
- Task 3: 1 error each
- Earbuds guidance was polarizing (2/3 disliked)

All tasks were completed (100% success), but clarity issues in Task 2 and Task 3 prevent the design from fully meeting our usability goals.

# ISSUES IDENTIFIED FROM TESTING

- Many users were unfamiliar with the newly introduced AI-generated beta feature and unsure how to use it
- Users confused creating an AI-generated beta with creating their own custom beta
- Users were unclear about the term "Injuries" in the body profile input section, mistaking it for injuries they wish to prevent rather than current injuries they have
- Some climbers do not prefer to wear earphones while climbing, indicating a need for adjustments in our complex tasks
- Users prefer to watch betas that are specific to the route they are attempting rather than generic betas

# PROGRESS TOWARDS THESE GOALS

## Efficiency

Based on usability testing, several design changes were implemented to reduce the time required to complete key tasks.

Terminology was clarified (e.g., “Generate AI Beta”) to reduce hesitation during personalization inputs.

We also improved step-by-step mode by adding clearer progress indicators and resume visibility, allowing users to quickly re-enter a climb without confusion.

Additionally, enhanced filtering and browsing features for community-shared betas help users locate relevant routes more efficiently between attempts.

## Clarity / Error Prevention

Testing revealed multiple mismatches between user expectations and system understanding, leading to navigation errors and control misinterpretation.

To address this, we refined interface terminology and added explanatory microcopy to clarify AI-generated beta creation and injury inputs.

System state visibility was improved in step-by-step mode through persistent progress indicators and clearer resume states, reducing situations where users became stuck or uncertain about next actions.

Finally, voice guidance was changed from a default interaction to an optional modality, minimizing confusion and supporting clearer interaction during climbing.

# NECESSARY FUTURE CHANGES

## Efficiency

Future improvements should aim to further reduce interaction burden during real climbing situations. While the current system provides pre-generated betas, it does not yet adapt to users' real-time movements. Future exploration could include recognizing users' movements in real time and delivering dynamically adapted beta guidance through voice feedback based on the current attempt. Such adaptive guidance could provide immediate feedback without requiring manual interaction, reducing interaction time and enabling a more efficient experience that does not interrupt climbing flow.

## Clarity / Error Prevention

Future work should continue focusing on reducing the gap between users' mental models and their understanding of the AI-driven system. Testing revealed that users tended to interpret personalized beta generation differently from their prior experiences with community-centered climbing apps. Therefore, beyond simple terminology changes, future iterations should develop onboarding and explanatory structures that help users progressively understand how AI-generated betas are created and how they differ from traditional community betas. In addition, system feedback and state visibility in step-by-step mode should be further strengthened so users can clearly understand their current position and next action without confusion.

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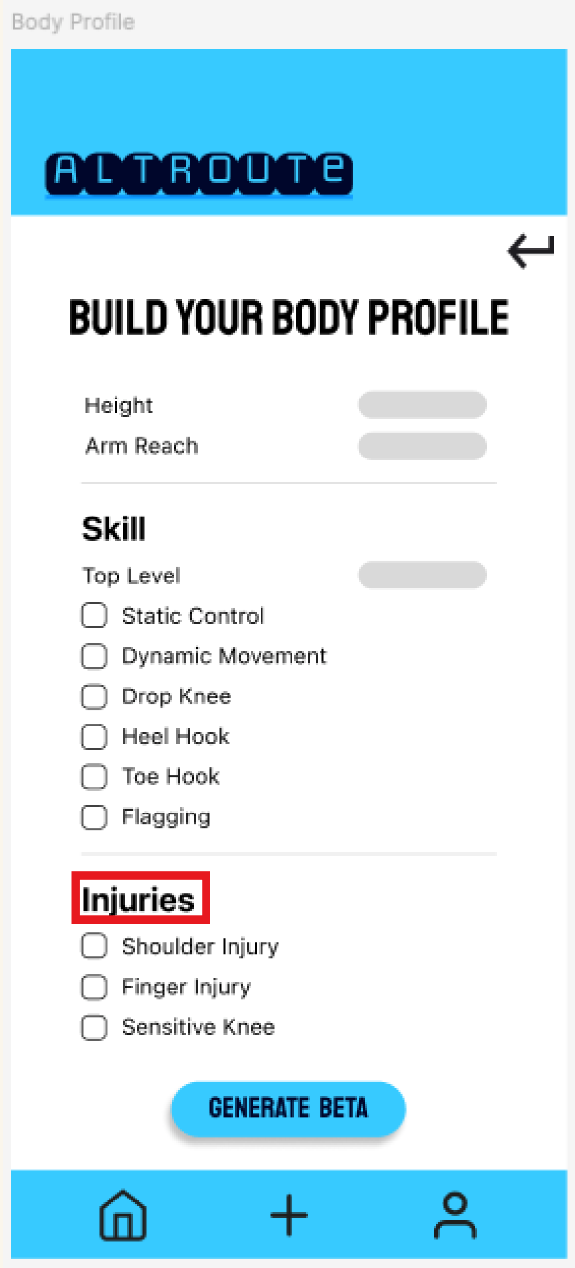
# 3 BIGGEST REVISIONS

# CHANGES

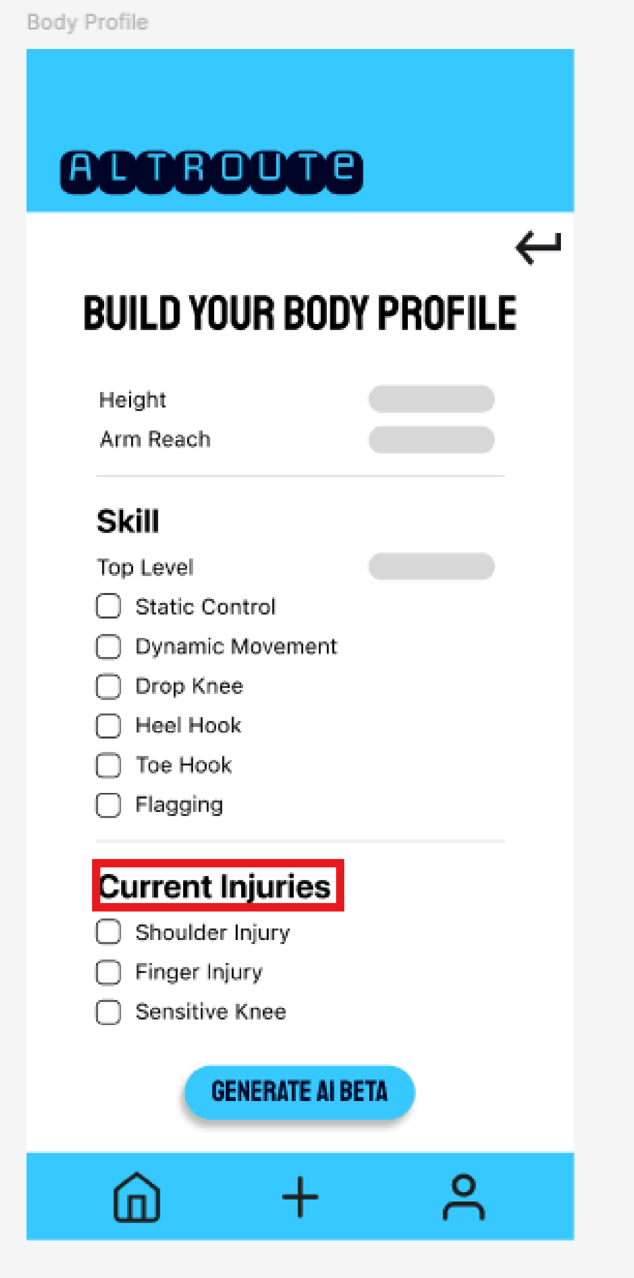
| Design Change | Efficiency (Time ↓) | Clarity / Error Prevention (Errors ↓) | Why it Supports the Goal |
|---|---|---|---|
| **Terminology clarification** (“Generate AI Beta”, “Current Injury”) | ✅ | ✅✅ | Prevents control misinterpretation and reduces hesitation during input |
| **Step-by-step system state visibility** (progress indicator, resume clarity) | ✅ | ✅✅ | Prevents users from getting stuck and enables faster task continuation |
| **Voice interaction as optional modality** | ⚠️ / ✅ | ✅ | Reduces confusion caused by unwanted audio interaction and supports safer interaction choices |
| **Community beta browsing improvements** (filters, friend betas) | ✅✅ | ✅ | Speeds up beta discovery and aligns with existing climbing app mental models |

# CHANGE #1 - TERMINOLOGY CLARIFICATION ("INJURIES" → "CURRENT INJURY")

Before




After




During usability testing:

- Some users interpreted "Injury" as injuries they want to prevent rather than their current condition
- Participants paused when filling out the body profile due to unclear wording

We changed:

- Change "Injury" to "Current Injuries", making it clear that users can specify their injuries to minimize strain when generating this beta

# CHANGE #1 - HOW IT PROGRESSES TOWARDS OUR USABILITY GOALS

## Efficiency

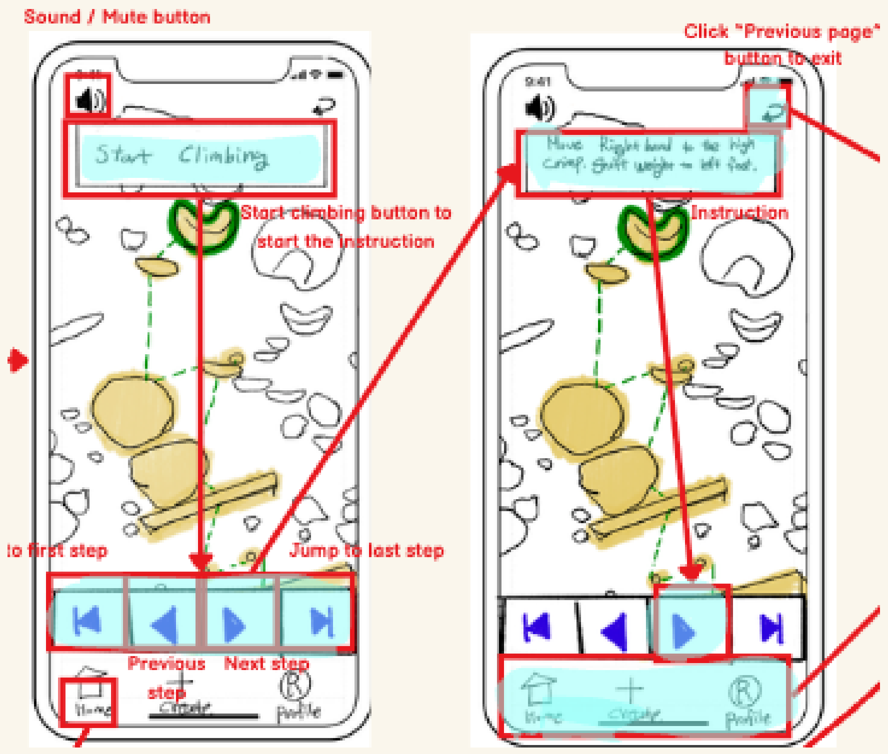
- Clearer wording helps users fill in their body profiles without hesitation, making the process faster
- Users would not be confused about what "injury" refers to

## Clarity / Error Prevention

- "Current Injuries" makes it obvious we're asking about active issues
- Fewer misclicks caused by unclear terminology

# CHANGE #2 - STEP-BY-STEP SYSTEM STATE VISIBILITY (PROGRESS INDICATOR, RESUME CLARITY)

## Before




During usability testing, several users:

- Forgot which step they were currently on
- Restarted the flow instead of resuming because they exited the beta
- Hesitated because system status was unclear

## After


- Stronger visual highlighting of the current step
- Improved transparency of system status to reduce uncertainty during climbing

# CHANGE #2 - HOW IT PROGRESSES TOWARDS OUR USABILITY GOALS

## Efficiency

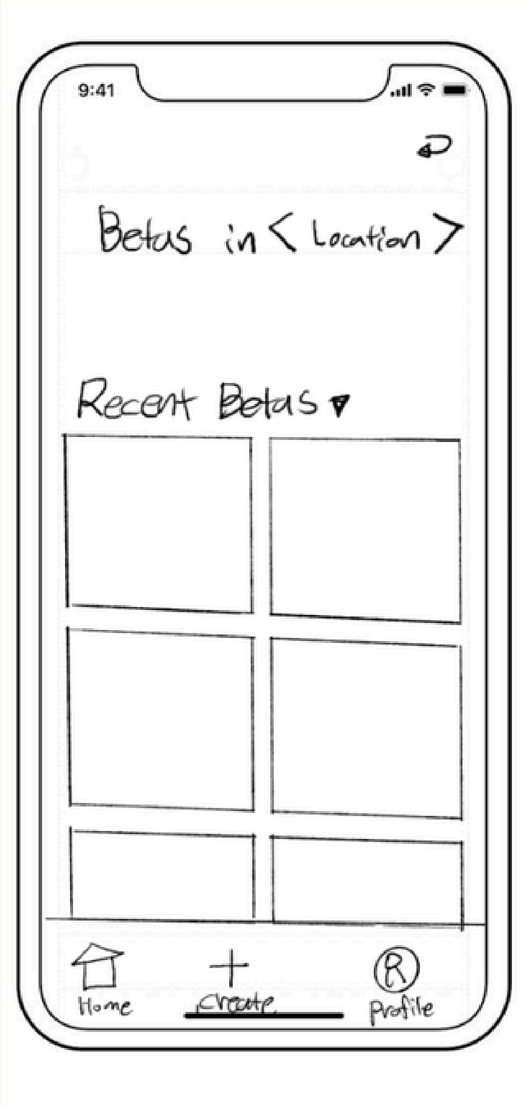
- Users can quickly identify their current position in the sequence
- Resume feature reduces time spent restarting the flow
- Clear step indicators reduce unnecessary back-and-forth navigation

## Clarity / Error Prevention

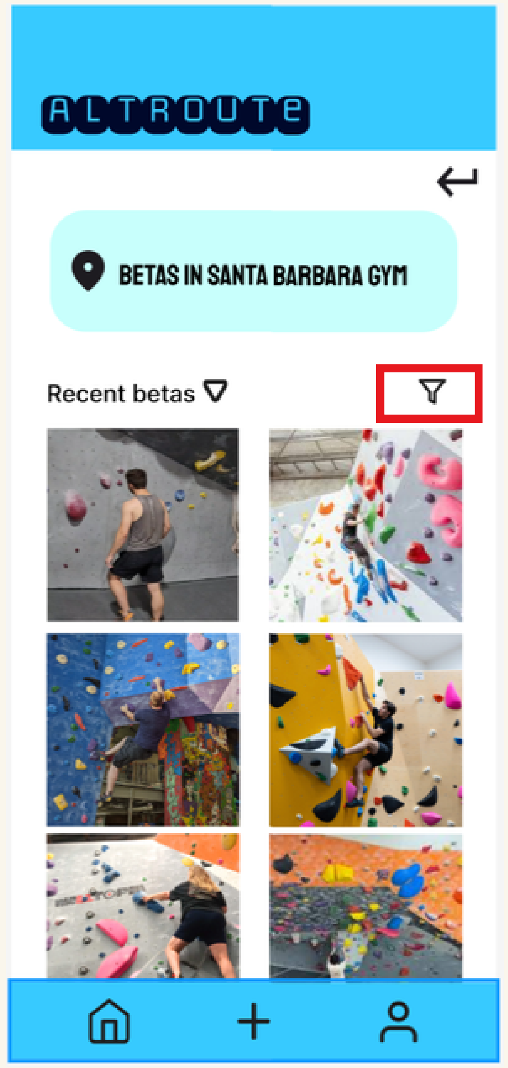
- Reduces confusion about system status
- Prevents accidental restarting of the beta sequence
- Makes next action immediately visible

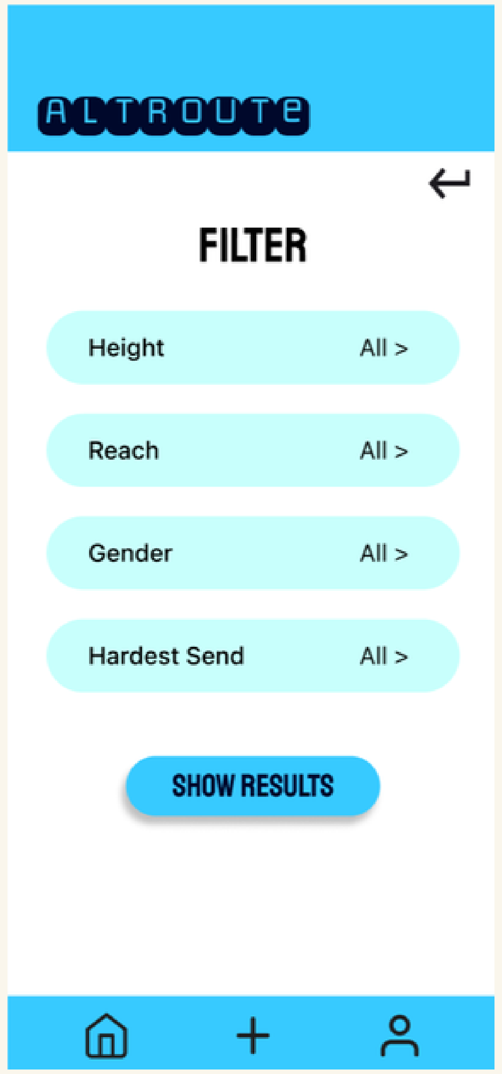
# CHANGE #3 - COMMUNITY BETA BROWSING IMPROVEMENTS (FILTERS, FRIEND BETAS)

**Before**




**After**




During usability testing:

- Users scrolled through many irrelevant betas before finding a relevant one
- Participants wanted to filter by height, skill level, or route type
- Some users want to find friend's beta

We changed:

- Introduce a community feature for users to share and find their friends' betas
- Introduce a filter to refine the search for community-shared betas based on desired route and body type

# CHANGE #3 - HOW IT PROGRESSES TOWARDS OUR USABILITY GOALS

## Efficiency

- It's faster to find the beta you're looking for through filters than scrolling through everything
- Filters help narrow things down, so users don't waste time checking irrelevant posts
- Seeing friends' betas by using filters!

## Clarity / Error Prevention

- Filters make the browsing experience less overwhelming
- Users are less likely to click into betas that don't fit their height, reach, or climbing level

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# MEDIUM-FI TASK FLOWS

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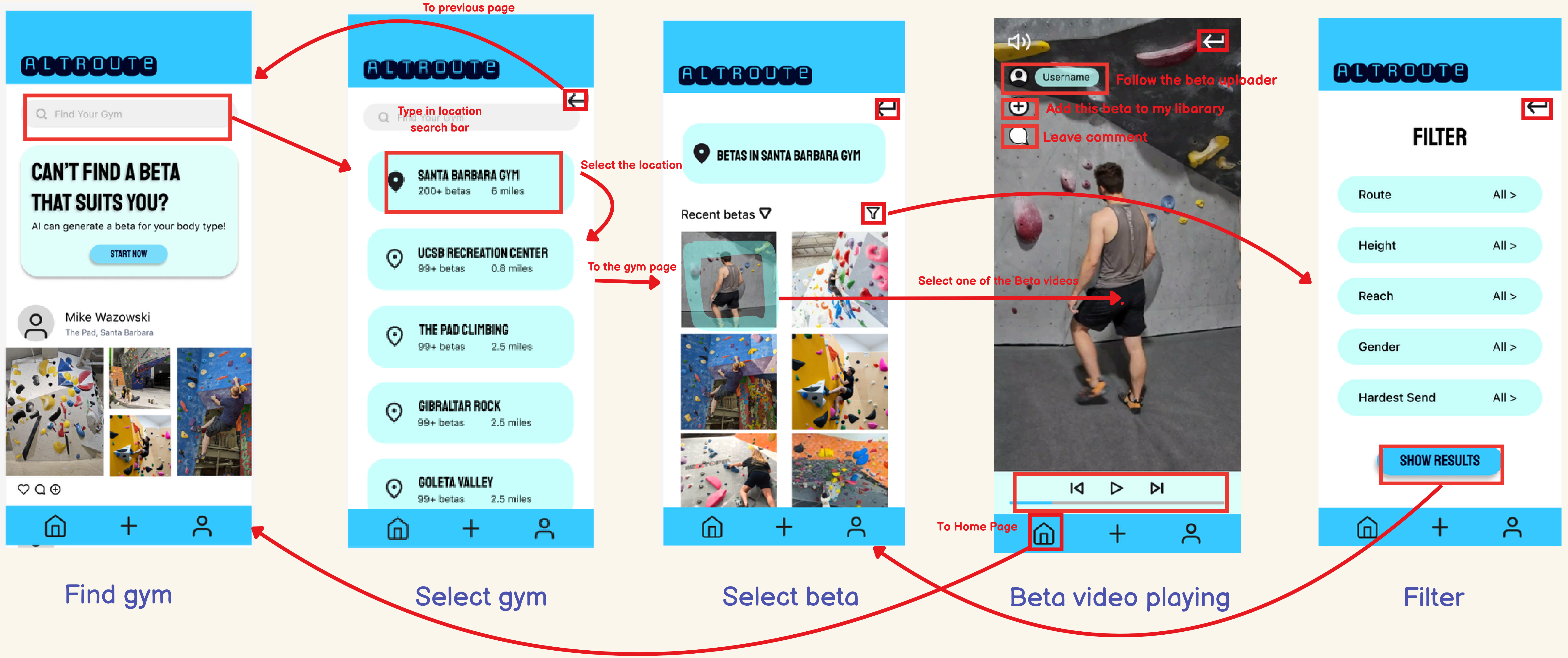
SCREEN SKETCHES

# SIMPLE: Users can search for a route-specific beta by searching for the gym and filtering the betas.

Start:




End

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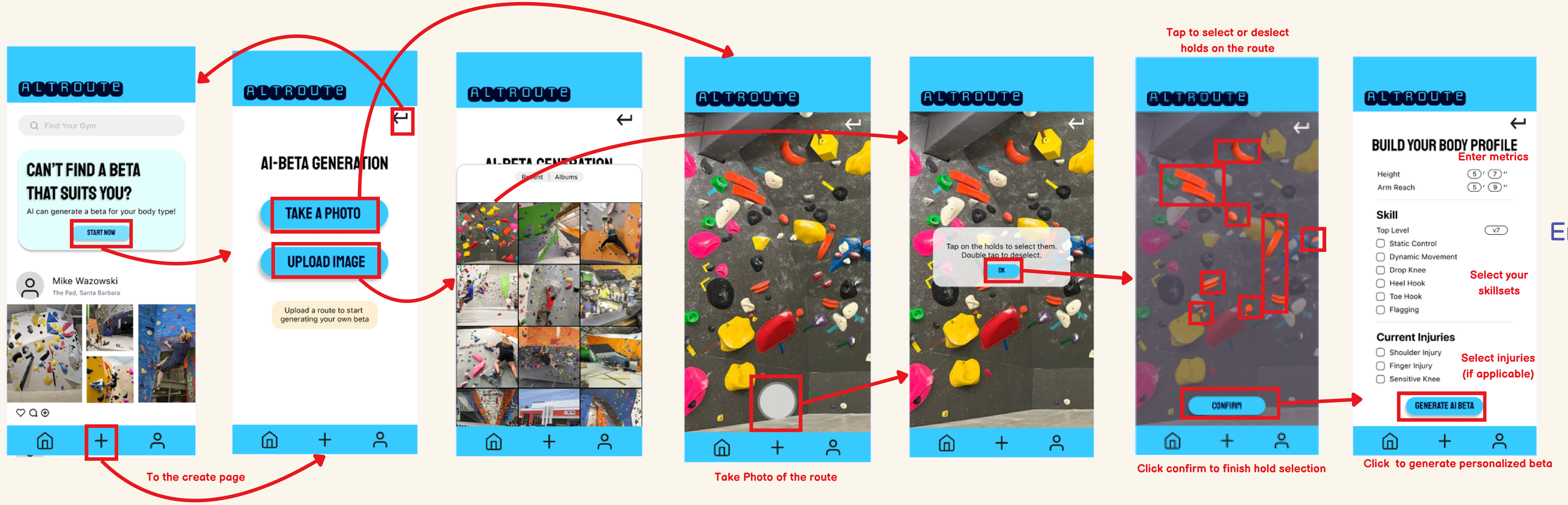
# MODERATE: Users can enter route details, physical attributes, and skill level to create personalized beta

Start:




End

Create beta | Upload route | Select image or Take photo | Select instructions | Select holds | Input and generate

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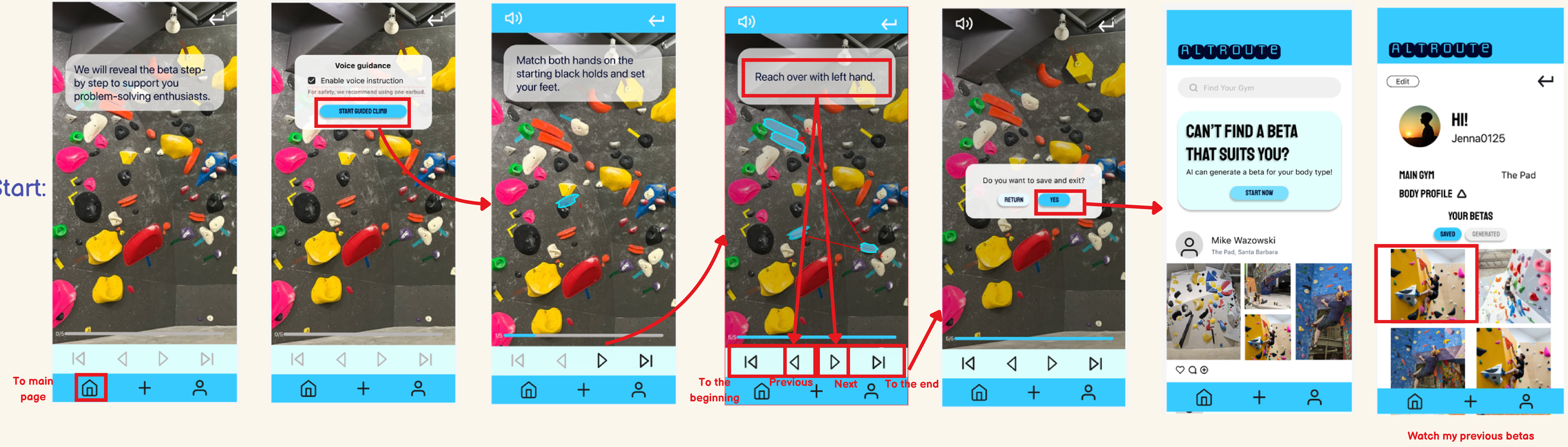
SCREEN SKETCHES

# COMPLEX: Users receive real-time guidance from the AI voice coach or visual guides for specific steps in the climb

Start:




End

Create beta instructions | Voice option | AI Beta step1 ............ AI Beta step5 | Save Beta | Return to Home | Profile (scroll)

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# PROTOTYPE IMPLEMENTATION

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# FIGMA

## Pros:

- Easy to iterate quickly after feedback
- Simple to connect screens and create multiple flows
- Works well for mobile layout and spacing
- No other installation needed

## Cons:

- No real backend logic or AI pr
- Personalization is simulated
- Limited realism for video/voice interactions
- Designed not real data
- Hard to represent real-time system responses

# LIMITATION

- Real AI-based dynamic beta generation was not implemented
  - Replaced with a hard-coded approach
  - Intended to validate interaction understanding and user responses before investing in full technical implementation
- Voice guidance was not implemented
  - Audio playback and interaction were not supported within the Figma prototyping environment
- Hard-coded community beta postings to simulate searching and browsing for betas
- Embodied interaction testing in a real climbing environment was limited
  - Testing was conducted in a simulated setting, which could not fully capture physical and environmental factors present during real climbing
- Unable to accurately depict selecting and deselecting holds for the moderate task in Figma prototyping environment, currently selects and deselects only in a specific order
- Unable to implement "Resume" feature as explained on Slide 30 in Figma prototyping environment. Users are supposed to be able to resume where they last stopped after exiting the beta viewing mode.

# WIZARD OF OZ + HARD CODED FEATURES

- AI beta generation is pre-designed (output does not change)
- Body profile fields do not affect the result (result is fixed)
- Gym list and betas are sample data
- Save / Comment actions are visual only
- Step-by-step guidance is pre-written, not dynamically generated
- Voice guidance is simulated and not interactive

# IMPACTS

- Users can evaluate the flow and clarity, but not AI accuracy
- Personalization feels realistic, but it is not adaptive
- Community features show structure, but not real social interaction
- Voice guidance shows intended use, but does not respond to user behavior
- Testing mainly reflects navigation and usability, not system intelligence

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# APPENDIX

CLIMBSPACE

# CLIMBING TERMS

**SEND**
A "SEND" IS ACHIEVED WHEN A CLIMBER FINISHES A CLIMB WITHOUT FALLING, REGARDLESS OF WHETHER IT IS THEIR FIRST TRY OR THEY HAVE PRACTICED IT BEFORE.

**ROUTE/PROBLEM**
A SPECIFIC CLIMBING PATH SET ON THE WALL USING DESIGNATED COLORED HOLDS.

**BETA**
A CLIMBER'S STRATEGY OR SEQUENCE OF MOVES FOR FINISHING A ROUTE.

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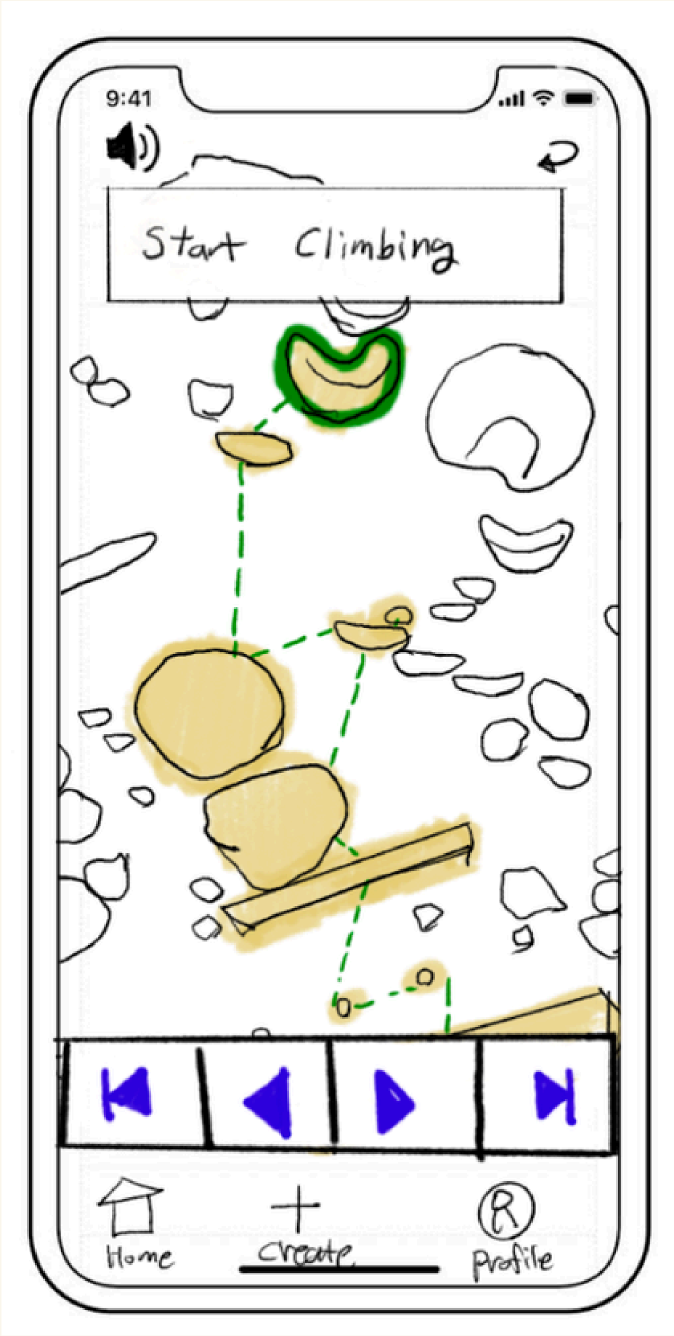
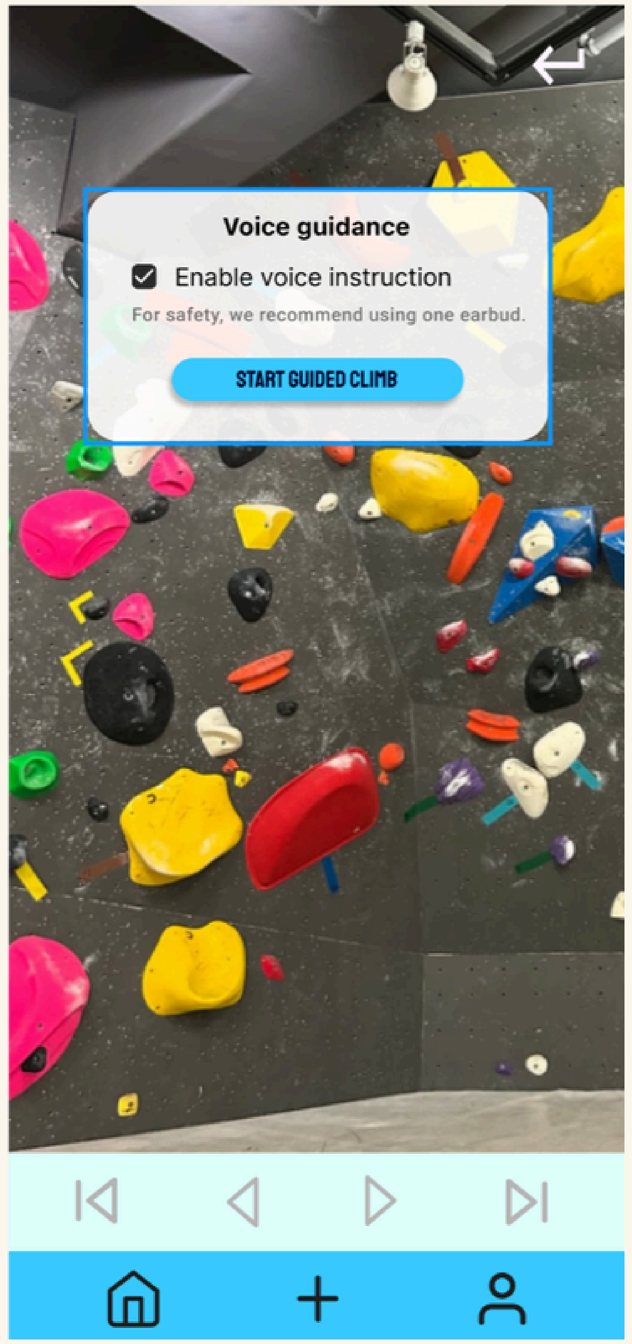
# CHANGE #4 -VOICE INTERACTION AS OPTIONAL MODALITY

Before




After




During usability testing:

- Most participants reported they would not wear earbuds while climbing, whether it be for safety or comfort reasons
- Some users felt uncomfortable playing audio instructions in a public gym

We changed:

- Make voice/earbuds feature optional and provide a clear visual alternative for alternative visual step-by-step beta

# CHANGE #4 - HOW IT PROGRESSES TOWARDS OUR USABILITY GOALS

## Efficiency

- It's quicker to start climbing since users don't have to set up earbuds first (now optional)
- The visual steps are easy to glance at between attempts
- It makes the flow feel more natural during practice

## Clarity / Error Prevention

- Users won't miss instructions if they can't hear the audio clearly.
- The on-screen guidance makes each step clearer.
- It avoids awkward situations where voice instructions play loudly in a public gym.

# LINK TO DELIVERABLES

**FIGMA:**
HTTPS://WWW.FIGMA.COM/DESIGN/MM2RDZLNWCBYRYNOD0TPXQ/ALTROUTE?NODE-ID=0-1&P=F&T=UGCGY1O2AXOKZABH-0

**FIGMA PROTOTYPE:**
HTTPS://WWW.FIGMA.COM/PROTO/MM2RDZLNWCBYRYNOD0TPXQ/ALTROUTE?NODE-ID=0-1&T=CCM0V3KG4W2YWC03-1

**README:**
HTTPS://DOCS.GOOGLE.COM/DOCUMENT/D/1PHGDVPCLWRCRVRB7N30KXYMQSVAYYI6RCHHQXFYR-78/EDIT?TAB=T.0